Yuan Jack Yao

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Education

Georgia Institute of Technology **Aug 2025 – Dec 2026**

M.S. in Analytics @ School of Computational Science and Engineering *Atlanta, Georgia*

- CS/CSE Courses: Conversational AI, Natural Language, Machine Learning, ML in Computational Biology

National Taiwan University **Sep 2014 – Jun 2018**

Bachelor's Degree in Economics *Taipei, Taiwan*

Work Experiences

Georgia Institute of Technology **Aug 2026 – Dec 2026**

CS 6604 Conversational AI Graduate Teaching Assistant *Atlanta, Georgia*

- **Designed & deployed assignments** for an 80-student course: **LoRA SFT + DPO** with unit-test autograder; **RAG with cross-session memory**; held weekly office hours to guide students for debugging

Walmart **Jun 2026 – Aug 2026**

Data Scientist III Intern @ Sam's Club AI Acceleration Team *Bentonville, Arkansas*

- Integrated **LLM & RAG with filtered vector search** into dashboard for camera/area/site-level metadata queries, with **cosine similarity & LLMs-as-a-Judge** against hallucinations, and an **embedding signature** verifying Milvus ingestion & retrieval use the identical embedding model, **eliminating silent retrieval degradation**
- Architected a **Cassandra-sourced** data layer feeding **PostgreSQL** and **Milvus**, regenerable after data loss
- Deployed Docker & Kubernetes CronJobs for **1,245 in-store cameras across 9 sites**: monthly Gemini Vision area classification with **full-fleet inference in 13 minutes** through high-throughput concurrency, glibc malloc arena caps and image pre-downsampling to **improve reliability by eliminating OOM**; hourly asynchronous snapshot capture to Azure, cutting **50% time** and **75% storage cost** via 7-day TTL

SUEZ **May 2022 – Sep 2024**

Junior Data Scientist @ Smart and Environmental Solutions (SES) China *Shanghai, China*

- Streamlined **NLP classifier** for **client's imbalanced** SCADA alarms, reaching **82% production accuracy**
- Designed a **DBSCAN anomaly detector** for **client's** air compressors predictive maintenance, achieving **70% production F1 score** and forecasting failures **two weeks in advance despite having only a single feature**

Personal Projects & Products

LinesShines (www.lines-shines.com): NFL Stats System, **iOS/Android/macOS/Windows Installable**

- Solely owned an installable NFL metrics PWA end to end: preprocessing pipeline, PostgreSQL ingestion, FastAPI JSON API, vanilla-JS frontend with Service Worker caching, deployed on Railway
- Designed pinnable multi-merging player comparison system with live-recomputing rank & percentile panels across multi-team/player union filters, inspired by GitHub's pinned repos & pull requests merge mechanics
- Rebuilt human-AI workflow on agent-skills-style layers: a CI-enforced pairing of pytest/Playwright tests with scope docs, a discussions-to-next-steps approval flow, and an always-loaded instruction file cut from 79 KB to under 3 KB

MonkeyJOB: Jack's Online Blackjack (www.jack-s-onlineblackjack.com): **Installable Online Game**

- Developed **full-stack Blackjack** with casino-style UI via FastAPI backend & vanilla JavaScript

Open Source

Pydantic AI (21k Stars, 3k Forks) — Agentic AI Framework — github.com/pydantic/pydantic-ai

- **Unified API errors handling pattern** across Google & Cohere embedding models by shared context managers

Code Puppy (870 Stars, 293 Forks) — Coding Agents — github.com/mpfaffenberger/code_puppy

- **Cut down idle CPU wake-ups by 88.5% for Agent-UI communications, restored silently-dropped user config across Gemini classes**, unified tools' drifted file-read process, and aligned 20+ inconsistent parser calls

LeetCode Wiki (36.6k Stars, 9.5k Forks) — Data Structures & Algorithms — github.com/doocs/leetcode

- Authored **8 optimal solutions with tutorials** for **hard** problems, including No.2454 in O(n) via double stacks

Technical Skills

Agents & ML: AI Infra & Platform, RAG, LLM Fine-tuning, Distillation, PyTorch, Hugging Face, TensorFlow, Keras
Programming & Databases: Python, SQL, C++, JavaScript, PostgreSQL, MySQL, Milvus, Cassandra, ETL
Web & Backend: FastAPI, Flask, REST APIs, OpenAPI, HTML, CSS, Plotly, Web Scraping
CI/CD, Tests & Cloud: CI/CD, pytest, Playwright, Linux, GitHub, Docker, Kubernetes, Azure, AWS, GCP